

Processando config 16, seed 0, teste: BlockAlphabit  
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 1.66  
p-value of test : 0.44

-----  
CPU time used : 00:00:00.03

Generator state:  
16777216 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 11.49  
p-value of test : 0.18

-----  
CPU time used : 00:00:00.03

Generator state:

33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 130.36  
p-value of test : 0.43

-----  
CPU time used : 00:00:00.01

Generator state:

50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32454.36  
p-value of test : 0.89

-----  
CPU time used : 00:00:00.04

Generator state:

67108864 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 161.13  
p-value of test : 0.80

-----  
CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 305  
Chi-square statistic : 242.57  
p-value of test : 0.9965

-----

CPU time used : 00:00:00.03

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

Normal statistic : 0.31  
p-value of test : 0.38

-----

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 32.58 |
| p-value of test              | : | 0.44  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 18.92 |
| p-value of test              | : | 0.96  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 41.49 |
| p-value of test              | : | 0.12  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 29.55 |
| p-value of test              | : | 0.24  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 17.78 |
| p-value of test              | : | 0.27  |

-----  
CPU time used : 00:00:00.06

Generator state:

134217728 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 64.75 |
| p-value of test              | : | 0.38  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 52.77 |
| p-value of test              | : | 0.79  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 166.93 |
| p-value of test              | : | 0.34   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 61.87 |
| p-value of test              | : | 0.31  |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 32     |
| ChiSquare statistic          | : | 61.81  |
| p-value of test              | : | 1.2e-3 |

-----  
CPU time used : 00:00:00.06

Generator state:  
150994944 bits have been read.

===== Summary results of Alphabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.31                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic            : 2.99
p-value of test                   : 0.22
```

```
-----
CPU time used                     : 00:00:00.03
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 8    |
| Value of the statistic       | : | 8.82 |
| p-value of test              | : | 0.36 |

-----  
CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216

Number of cells =  $2^L$  = 256

Expected number per cell = 65536

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 131.50 |
| p-value of test              | : | 0.40   |

-----  
CPU time used : 00:00:00.01

Generator state:

201326592 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 33159.22
p-value of test : 0.06
```

```
-----
CPU time used : 00:00:00.04
```

Generator state:

218103808 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring\_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 191.64
p-value of test : 0.21
```

```
-----
CPU time used : 00:00:00.01
```

Generator state:

234881024 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 338.64 |
| p-value of test              | : | 0.09   |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.01 |
|---------------|---|-------------|

Generator state:

251658240 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |       |
|------------------|---|-------|
| Normal statistic | : | -0.32 |
| p-value of test  | : | 0.63  |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.01 |
|---------------|---|-------------|

Generator state:

268435456 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 24.53 |
| p-value of test              | : | 0.82  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 32.96 |
| p-value of test              | : | 0.37  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 25.60 |
| p-value of test              | : | 0.78  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 26.95 |
| p-value of test              | : | 0.36  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 10.13 |
| p-value of test              | : | 0.81  |

-----  
CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 66.95 |
| p-value of test              | : | 0.31  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 74.61 |
| p-value of test              | : | 0.13  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 153.14 |
| p-value of test              | : | 0.64   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 45.79 |
| p-value of test              | : | 0.86  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 32.77 |
| p-value of test              | : | 0.43  |

-----  
CPU time used : 00:00:00.06

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufire_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.29                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1

-----
Test Results for Delta = 1.0000

Number of degrees of freedom      : 2
Value of the statistic            : 15.54
p-value of test                   : 4.2e-4 *****
```

```
-----
CPU time used                     : 00:00:00.01
```

```
Generator state:
318767104 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE

Number of bits = n = 16777216
```

Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 6.74  
p-value of test : 0.56

-----  
CPU time used : 00:00:00.03

Generator state:  
335544320 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 144.35  
p-value of test : 0.15

-----  
CPU time used : 00:00:00.01

Generator state:  
352321536 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32972.89  
p-value of test : 0.21

-----  
CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 184.58  
p-value of test : 0.33

-----  
CPU time used : 00:00:00.01

Generator state:

385875968 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 334.19  
p-value of test : 0.12  
-----

CPU time used : 00:00:00.01

Generator state:

402653184 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -1.15  
p-value of test : 0.88  
-----

CPU time used : 00:00:00.01

Generator state:

419430400 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 35.04 |
| p-value of test              | : | 0.33  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 23.34 |
| p-value of test              | : | 0.84  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 36.47 |
| p-value of test              | : | 0.27  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 22.82 |
| p-value of test              | : | 0.59  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 19.52 |
| p-value of test              | : | 0.19  |

-----  
CPU time used : 00:00:00.07

Generator state:

436207616 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 70.30 |
| p-value of test              | : | 0.22  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 63.07 |
| p-value of test              | : | 0.44  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 152.84 |
| p-value of test              | : | 0.64   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 59.04 |
| p-value of test              | : | 0.40  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 31.71 |
| p-value of test              | : | 0.48  |

-----  
CPU time used : 00:00:00.04

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

|                 |                                                |
|-----------------|------------------------------------------------|
| Version:        | TestU01 1.2.3                                  |
| Generator:      | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits: | 16777216                                       |

Number of statistics: 17  
Total CPU time: 00:00:00.26  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test                         | p-value |
|------------------------------|---------|
| -----                        | -----   |
| 1 MultinomialBitsOver, L = 2 | 4.2e-4  |
| -----                        | -----   |

All other tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells = 2^L = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:  
Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 2    |
| Value of the statistic       | : | 0.84 |
| p-value of test              | : | 0.66 |

-----  
CPU time used : 00:00:00.01

Generator state:  
469761792 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 8.02  
p-value of test : 0.43

-----  
CPU time used : 00:00:00.01

Generator state:

486539008 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 122.56  
p-value of test : 0.62

-----  
CPU time used : 00:00:00.01

Generator state:  
503316224 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32432.18  
p-value of test : 0.91

-----  
CPU time used : 00:00:00.03

Generator state:  
520093440 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 193.93  
p-value of test : 0.18

-----  
CPU time used : 00:00:00.03

Generator state:  
536870656 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 325.04  
p-value of test : 0.21

-----  
CPU time used : 00:00:00.01

Generator state:  
553647872 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.46  
p-value of test : 0.32

-----  
CPU time used : 00:00:00.01

Generator state:

570425088 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 46.25 |
| p-value of test              | : | 0.05  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 30.27 |
| p-value of test              | : | 0.50  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.50 |
| p-value of test              | : | 0.59  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 31.46 |
| p-value of test              | : | 0.17  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 14.49 |
| p-value of test              | : | 0.49  |

-----  
CPU time used : 00:00:00.06

Generator state:

587202304 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 67.17 |
| p-value of test              | : | 0.30  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 71.69 |
| p-value of test              | : | 0.19  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 174.21 |
| p-value of test              | : | 0.21   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 59.18 |
| p-value of test              | : | 0.40  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 28.28 |
| p-value of test              | : | 0.66  |

-----  
CPU time used : 00:00:00.04

Generator state:  
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L = 4$   
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----

Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 2    |
| Value of the statistic       | : | 2.39 |
| p-value of test              | : | 0.30 |

-----  
CPU time used : 00:00:00.01

Generator state:  
620756480 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic : 4.95
p-value of test : 0.76
```

-----  
CPU time used : 00:00:00.01

Generator state:

637533696 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic : 122.88
p-value of test : 0.61
```

-----  
CPU time used : 00:00:00.01

Generator state:  
654310912 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 33403.11  
p-value of test : 6.8e-3

-----  
CPU time used : 00:00:00.03

Generator state:  
671088128 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers  $\geq 10$

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 182.99  
p-value of test                   : 0.36  
-----
```

```
CPU time used                      : 00:00:00.01
```

Generator state:  
687865344 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0
```

Counters with expected numbers  $\geq 10$

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 281.75  
p-value of test                   : 0.83  
-----
```

```
CPU time used                      : 00:00:00.01
```

Generator state:  
704642560 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32
```

```
-----  
Normal statistic                   : -0.49  
p-value of test                   : 0.69  
-----
```

```
CPU time used                      : 00:00:00.01
```

Generator state:  
721419776 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 28.48 |
| p-value of test              | : | 0.65  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 26.61 |
| p-value of test              | : | 0.69  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 25.07 |
| p-value of test              | : | 0.80  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 19.27 |
| p-value of test              | : | 0.78  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 16.58 |
| p-value of test              | : | 0.34  |

-----  
CPU time used : 00:00:00.06

Generator state:  
738196992 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 64.86 |
| p-value of test              | : | 0.38  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 71.98 |
| p-value of test              | : | 0.18  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 167.26 |
| p-value of test              | : | 0.33   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 65.27 |
| p-value of test              | : | 0.21  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 40.04 |
| p-value of test              | : | 0.16  |

-----  
CPU time used : 00:00:00.04

Generator state:

754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 1.23  
p-value of test : 0.54

-----  
CPU time used : 00:00:00.01

Generator state:  
771751168 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 5.86  
p-value of test : 0.66

-----  
CPU time used : 00:00:00.01

Generator state:

788528384 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 111.57  
p-value of test : 0.85

-----  
CPU time used : 00:00:00.01

Generator state:  
805305600 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32807.88  
p-value of test : 0.44

-----  
CPU time used : 00:00:00.03

Generator state:  
822082816 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers  $\geq 10$

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 190.34  
p-value of test                   : 0.23  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

838860032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers  $\geq 10$

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 304.34  
p-value of test                   : 0.50  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

855637248 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : -0.30  
p-value of test                   : 0.62  
-----
```

-----  
CPU time used : 00:00:00.01

Generator state:  
872414464 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 26.79 |
| p-value of test              | : | 0.73  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 24.45 |
| p-value of test              | : | 0.79  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 37.41 |
| p-value of test              | : | 0.23  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 25.92 |
| p-value of test              | : | 0.41  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 10.53 |
| p-value of test              | : | 0.79  |

-----

CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----

Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 41.63 |
| p-value of test              | : | 0.98  |

-----

Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 67.63 |
| p-value of test              | : | 0.29  |

-----

Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 151.01 |
| p-value of test              | : | 0.68   |

-----

Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 78.81 |
| p-value of test              | : | 0.03  |

-----

Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 23.54 |
| p-value of test              | : | 0.86  |

-----

CPU time used : 00:00:00.04

Generator state:  
905968640 bits have been read.

===== Summary results of Alphabit =====

Version:            TestU01 1.2.3  
Generator:          ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits:     16777216  
Number of statistics: 17  
Total CPU time:     00:00:00.23

All tests were passed